

A presentation for delivering lecture
for course work of
SY BTech (All branches except civil engineering)
on



Essentials of Civil Engineering

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Unit 3

Building construction and building services

Course Outcomes: At the end of the course, the students are able to:
CO 1: Demonstrate different terminologies related to Civil Engineering.
CO 2: Select suitable structural component or structure, services in a particular situation.
CO 3: Apply the various concepts of environmental engineering in practice.
CO 4: Assess suitability of material or type of foundation in a particular situation.
CO 5: Choose provision of dam, weir, barrage or water conservation structures.

Unit 1	Introduction to Civil Engineering and Civil Engineering Materials [6Hrs]
	Introduction, Branches, Scope and Role of Civil Engineering, List of materials, Type, Properties, Uses of materials: Cement, Aggregate, Brick, Steel, Concrete, Stone, Soil, Mortar, Timber, Plastic, Fly ash, Steel slag, Bitumen, Pipe, Wire, Cable.
Unit 2	Building Planning and Transportation [7 Hrs]
	Principles of planning, orientation of buildings, introduction to 'Byelaws' regarding building line, height of building, open space requirements, F.S.I., Setbacks, ventilation, sanitation. Role of transportation in national development, Types of roads: Introductions to NH, SH, MDR, ODR, VR, and Expressway. Cross section of road, Transportation system- BRTS, Metro. Railway: Introduction, Gauges, Cross section.
Unit 3	Building Construction and Building Services [7 Hrs]
	Type of building, Component of building and its functions, types of load acting on building, types of brick bonds, typical building layout, Symbol used for water supply, plumbing, and sanitation. Types of building services like plumbing and sanitation, electricity, building finishes.
Unit 4	Foundation for Structures [6 Hrs]
	Introduction to foundation, Types of foundation and their suitability, Soil type as foundation strata, Concept of bearing capacity, ultimate, safe and presumptive bearing capacity, Shallow foundation and Deep foundations, Modes of failure.

TYPES OF BUILDING

Types Of Building

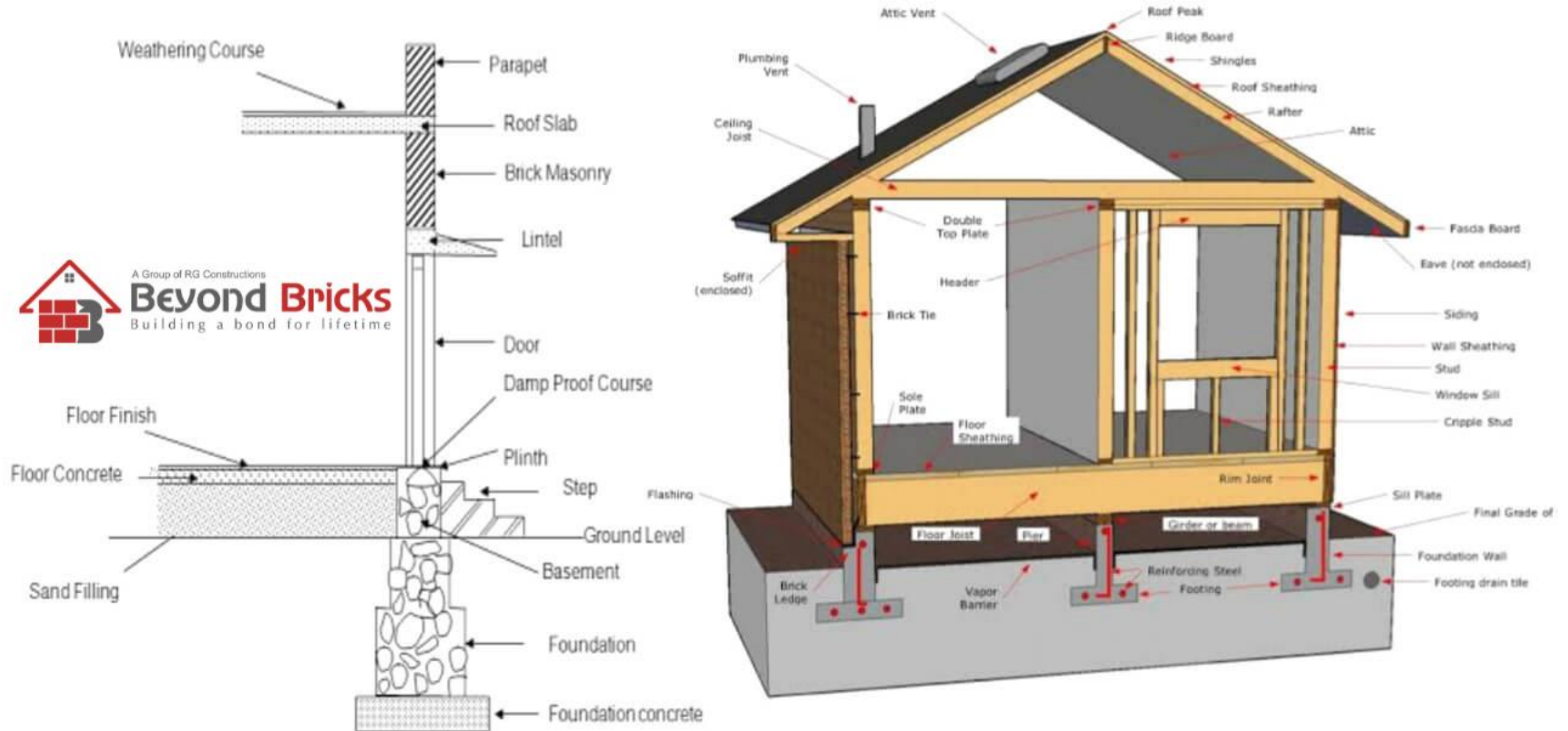
1. Agriculture Building
2. Commercial Building
3. Residential Building
4. Educational Building
5. Government office Building
6. Industrial Building
7. Military Building
8. Religious Building
9. Transport Building



Basic components of a Building



A Group of RG Constructions
Beyond Bricks
Building a bond for lifetime



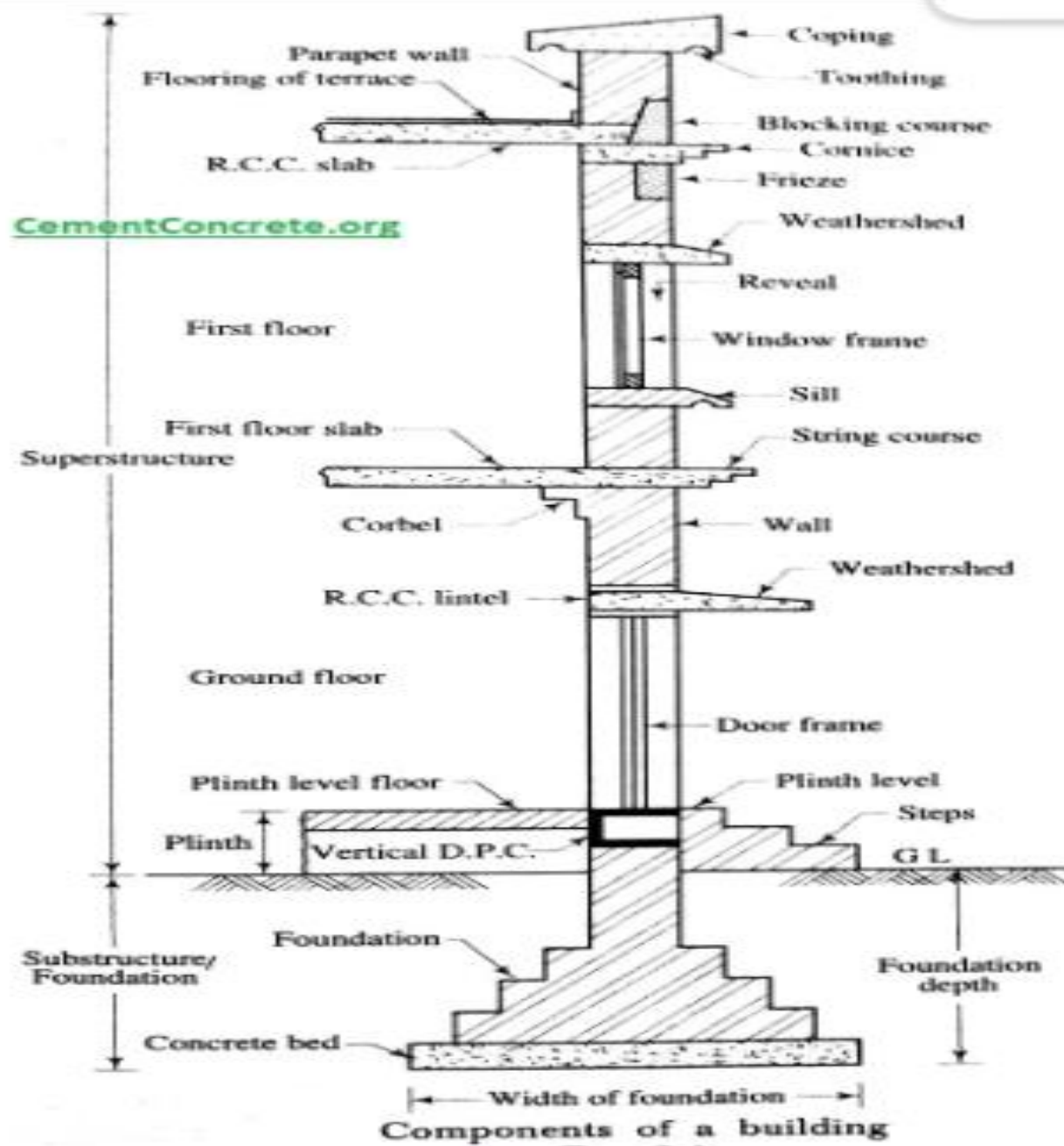
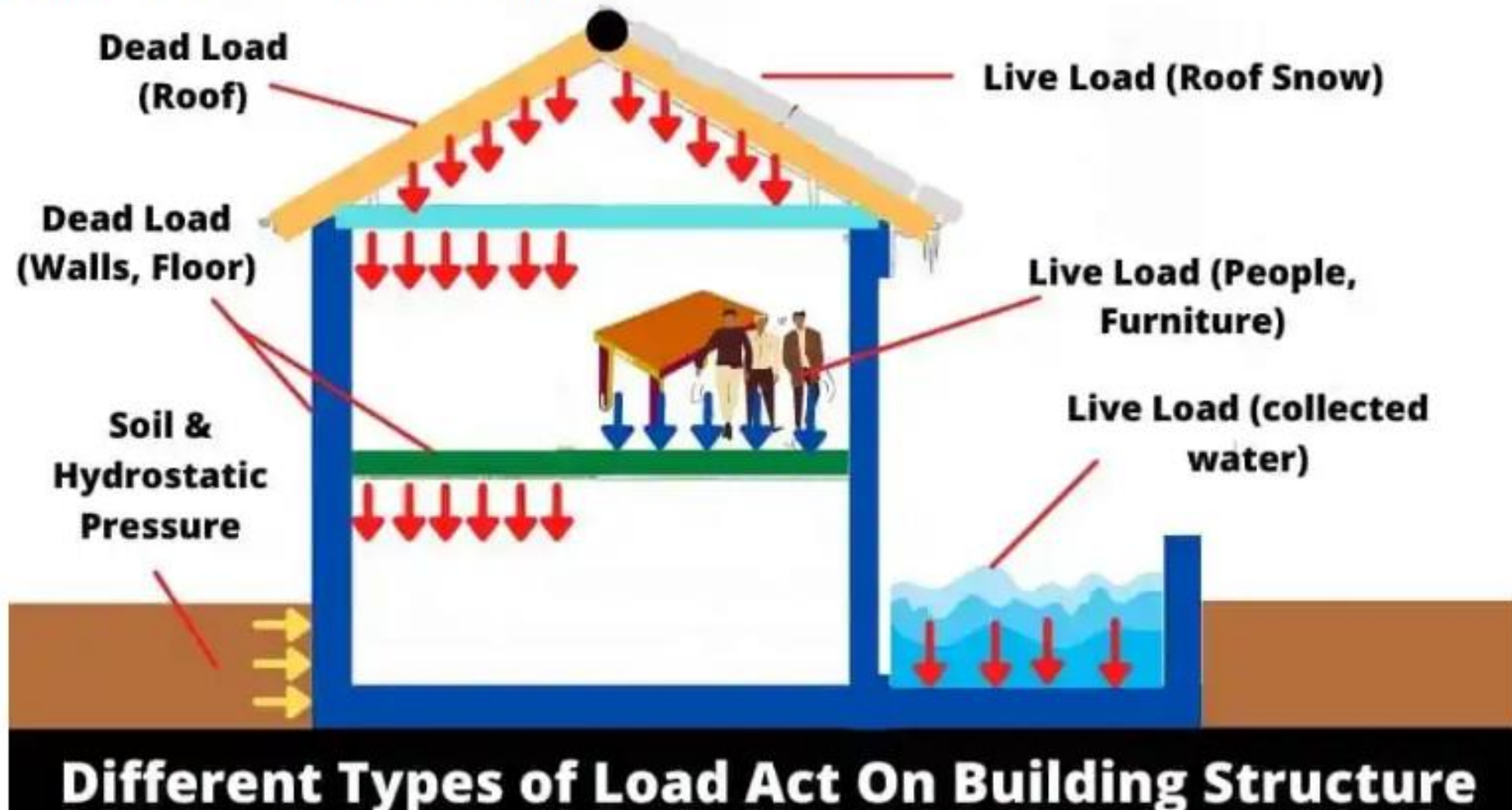


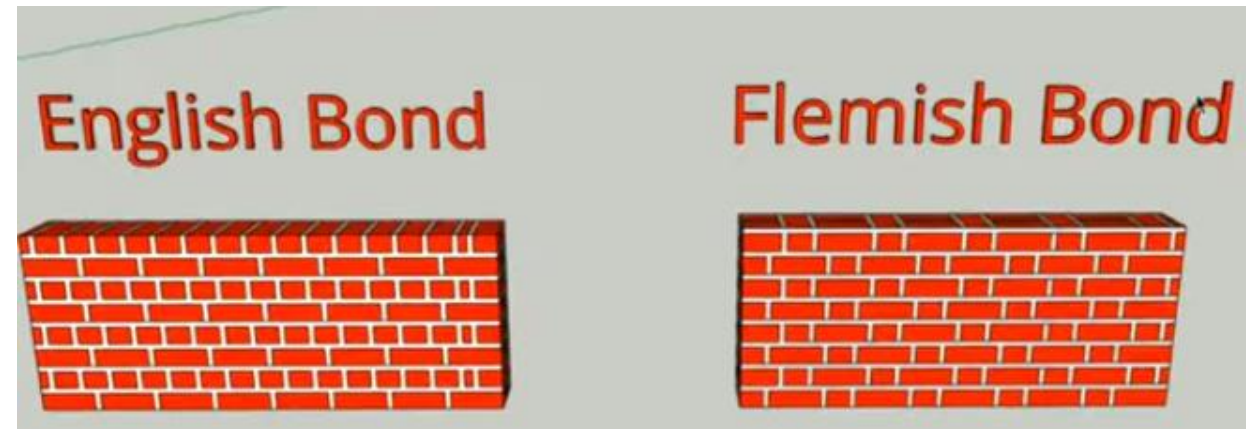
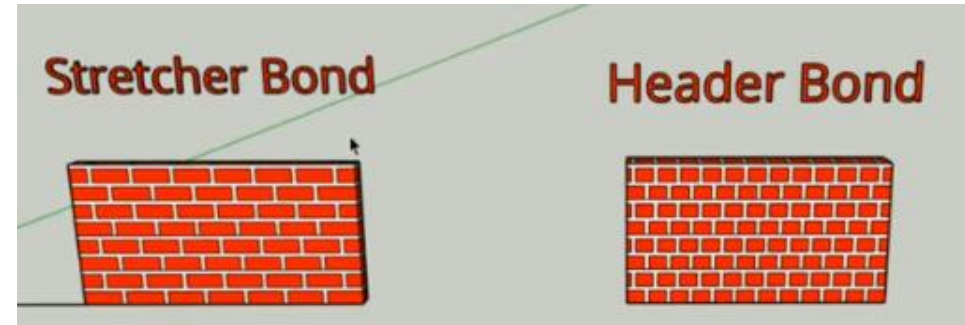
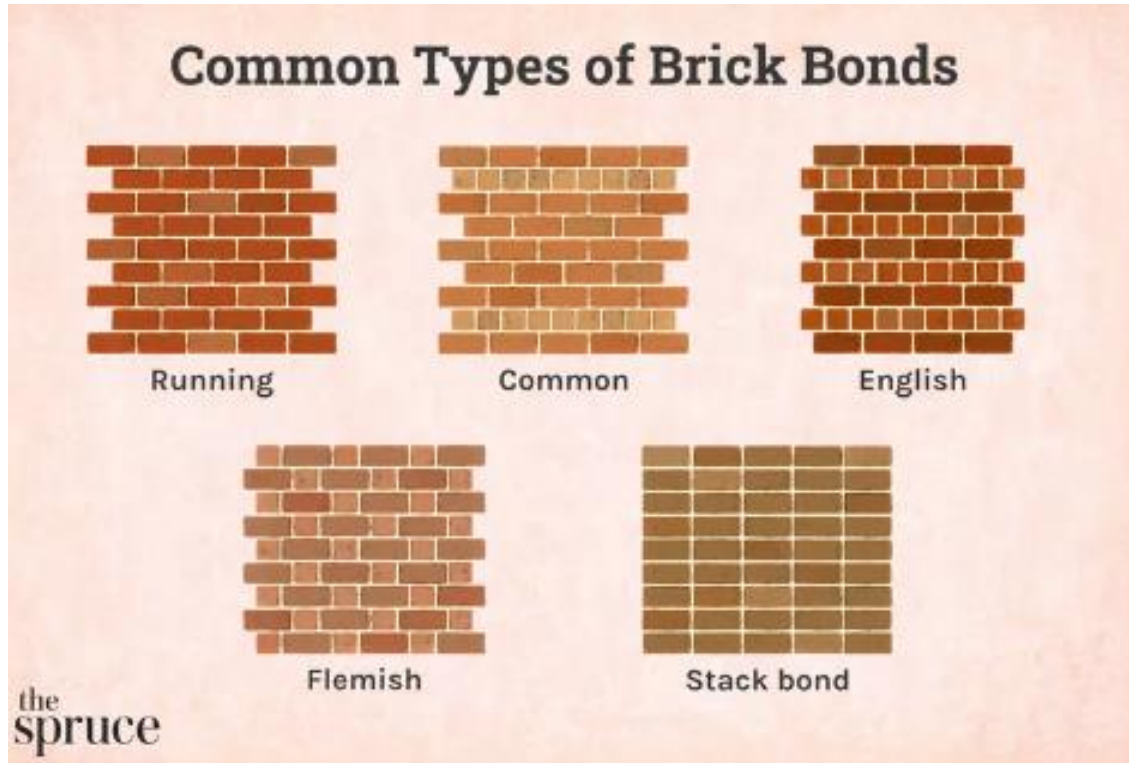
Fig.1. Important Building Components

TYPES OF LOAD ACTING ON BUILDING

[Home](#) » Types of Load On Building



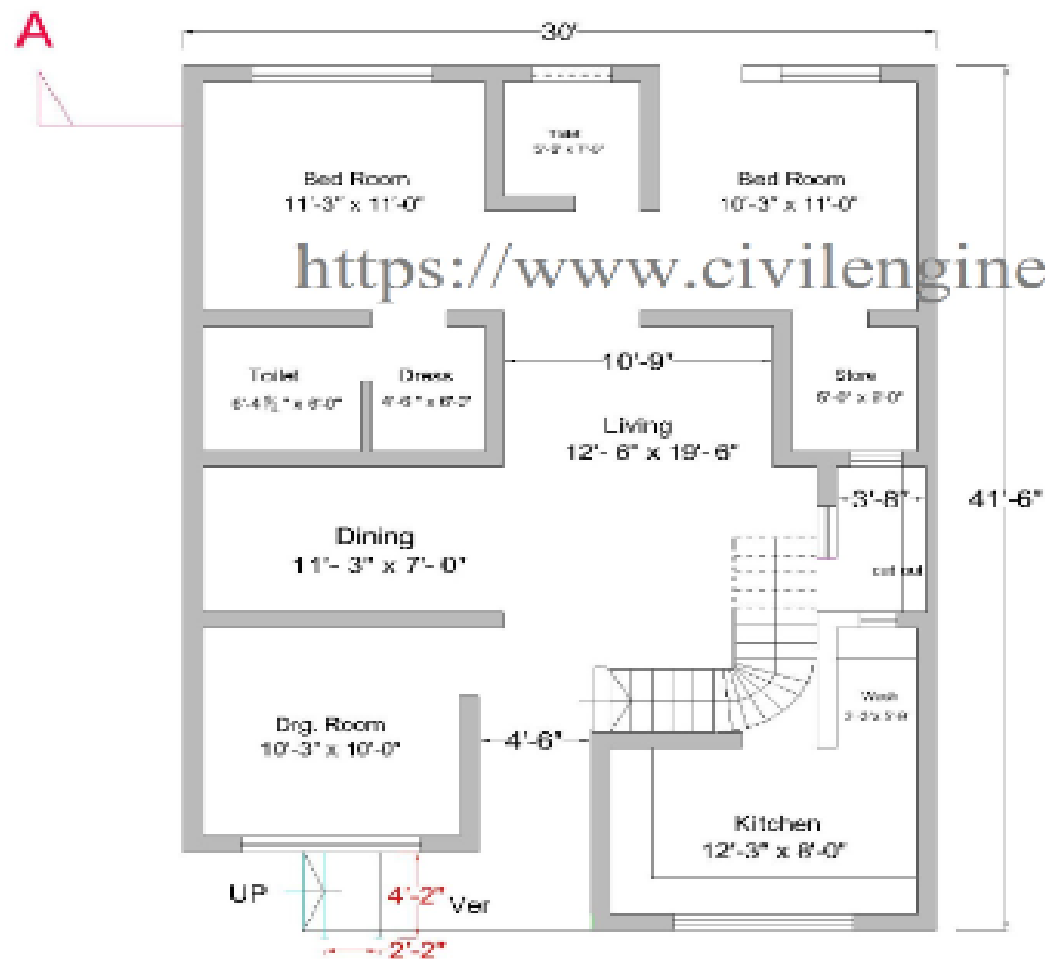
TYPES OF BRICK BOND



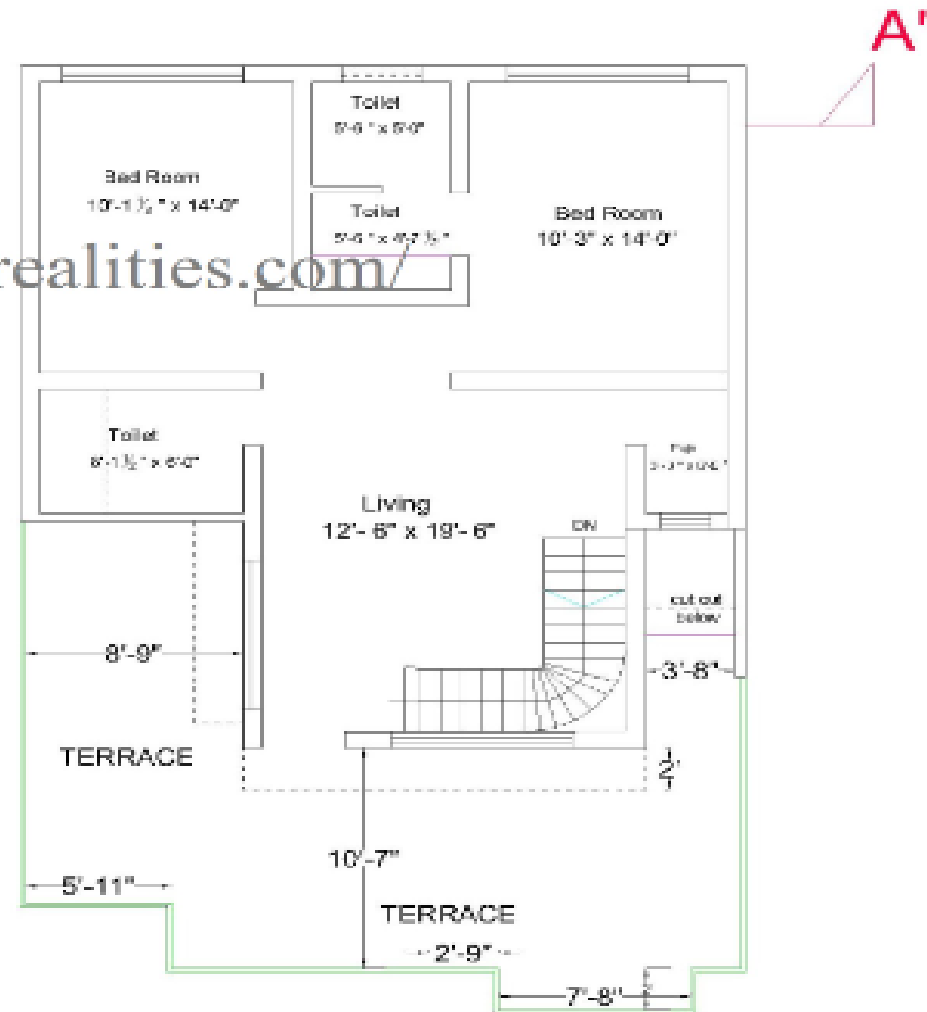
Comparative merits and demerits of English and Flemish bond

- Sushil kumar 236-237
- Entire bonds in brick work topic from Building Construction-Sushil Kumar

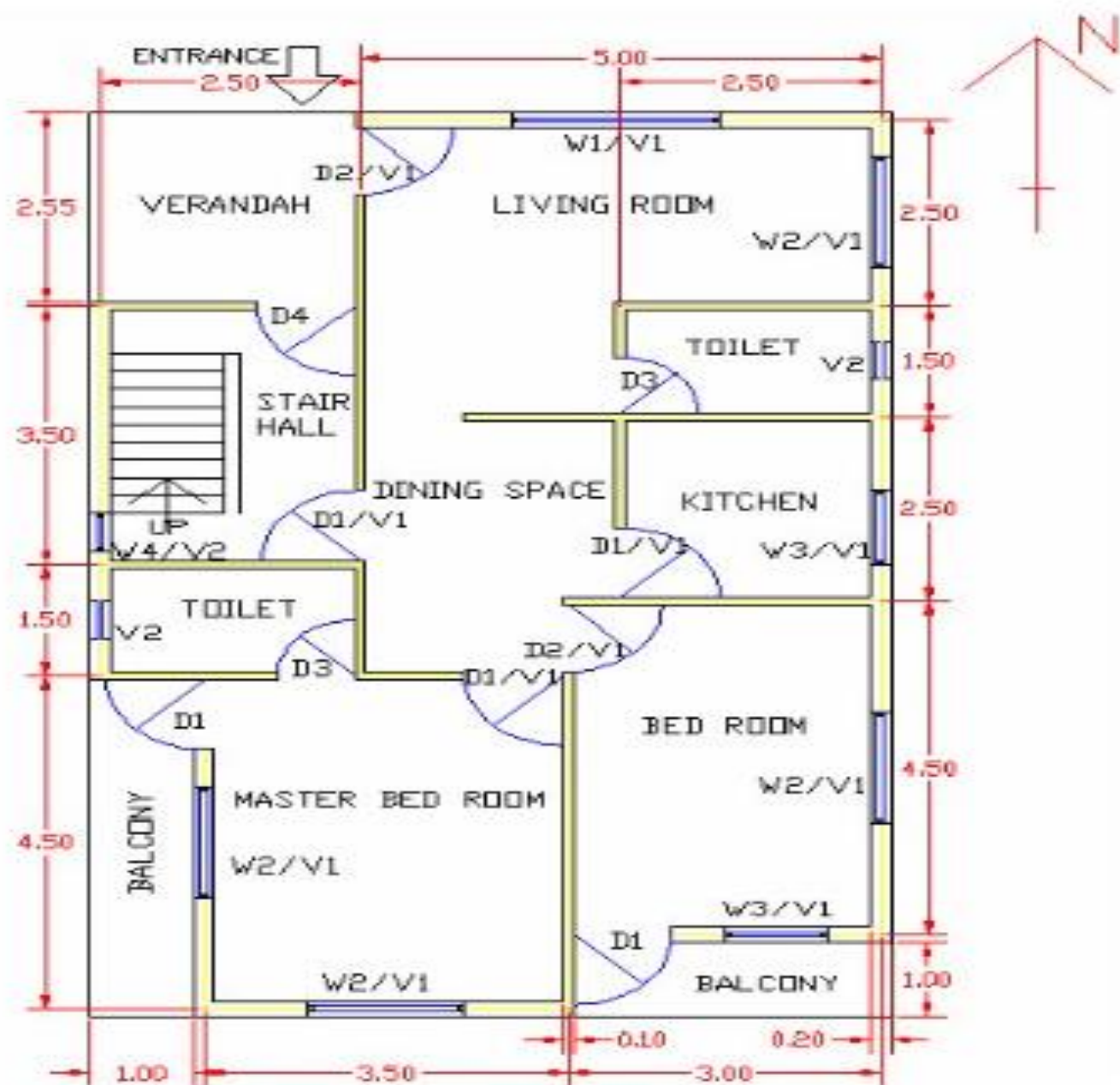
TYPICAL BUILDING LAYOUT



GROUND FLOOR PLAN



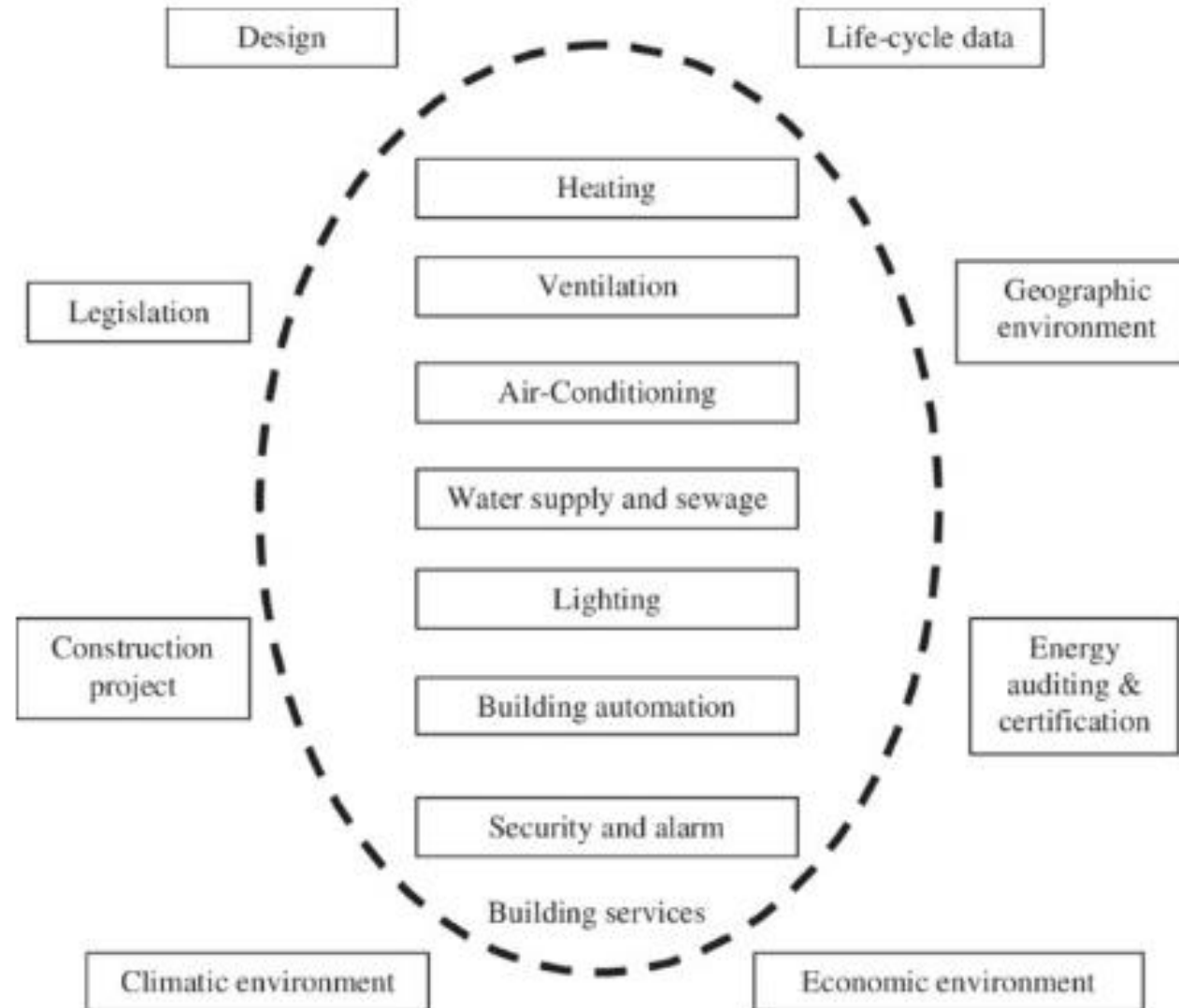
FIRST FLOOR PLAN





SYMBOL USED FOR WATER SUPPLY, PLUMBING AND SANITATION

TYPES OF BUILDING SERVICES



PLUMBING AND SANITATION

PLUMBING AND SANITATION WORK ITEMS USED IN BUILDING CONSTRUCTION

How do plumbers work?

A house sanitation cannot be complete without plumbing and sanitary arrangements. Regular and adequate supply of water and an efficient system for the disposal of the waste is a must for every residential Plumbing and sanitary system is necessary for every housing project, big or small.

Proper planning and designing of the plumbing and sanitary system serves the hygienic requirements of its occupants. Plumbing and sanitation is, therefore, an important branch of building technology. Approximately, 8% of the total building construction cost of the building is devoted to plumbing and sanitation works.

- Water supply.
- Soil pipes and fixtures.
- Sanitary drainage system, to carry the waste water from the plumbing fixtures to the public or private disposal system.
- Storm water drainage system to collect and carry rain water.
- Fire fighting systems for high rise buildings as per the statutory provisions. (All buildings may not need this)

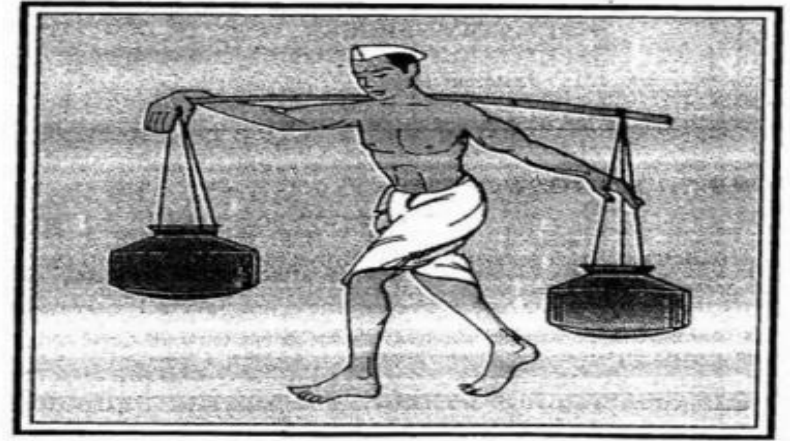


Fig. No. 29.1 Need of water supply system.

Water should be supplied with the required pressure, up to O.H.W.T. (Over Head Water Tank) and then to the individual residential units with the minimum prescribed pressure.

PRINCIPLES AND NORMS FOR PLUMBING AND SANITATION WORK

(1) WATER SUPPLY SYSTEM

Adequate quantity of clean and potable water for drinking purpose should be supplied. Water from other sources should also be made available for daily use, for twenty four hours efficient flushing.

Proper pumping systems with adequate stand by arrangements should be made.

The system should invoke minimum maintenance.

(2) PLUMBING FIXTURES AND PIPES

The plumbing fixtures should be -

Smooth finished for an uninterrupted flow.

Located in well ventilated enclosures/ducts.

Easily accessible for the intended use and repairs.

Able to withstand designed pressure.

Connected to a drainage system with water seal traps.

Tested for leakages, defects etc.

The plumbing pipes should be-

Made of durable material and connected by suitable and satisfactory joints of good workmanship, for satisfactory service during its reasonable life expectancy.

Easily accessible for inspection, working and repairs.

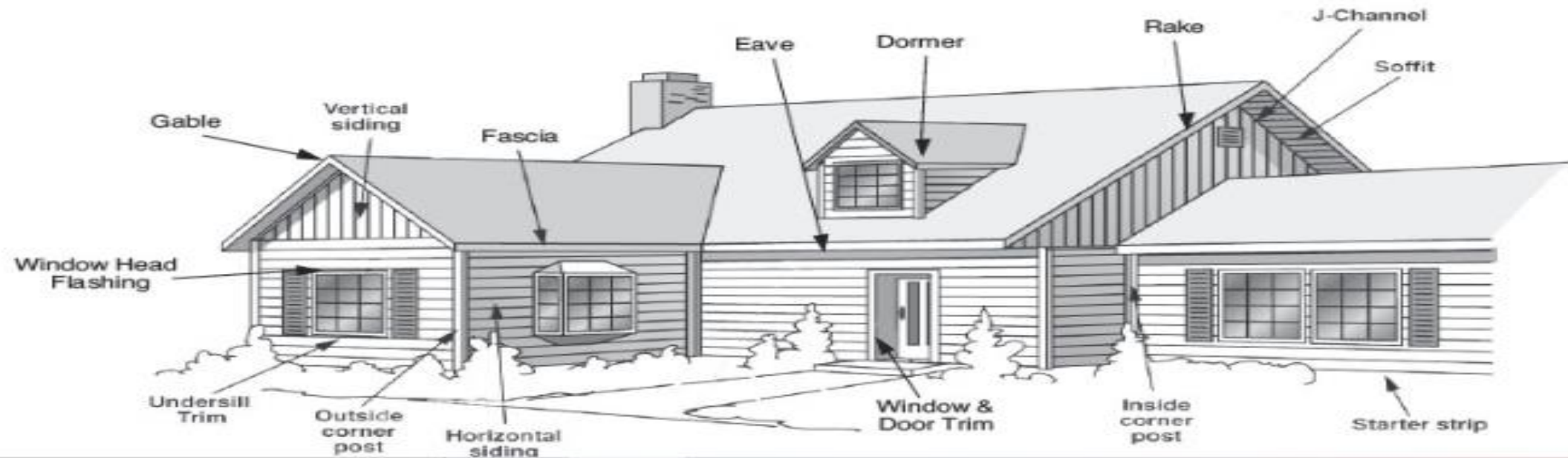
ELECTRICITY AND BUILDING FINISHES

4. Roof Finishes Types

- Corrugated steel
- Profiled light aluminium
- Wood shingles
- Concrete tiles
- Clay tiles
- Glazed
- Lightweight metallics
- Synthetic slates/tiles
- Natural slates
- Artificial mineral fibre
- Cement slates
- Titanium-cladding panels

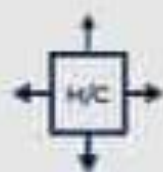
2. Wall Finishes

- Plastering
- Tiling
- Skirting
- Wallpaper
- Painting
- Dry Linings
- Molding
- Architraves

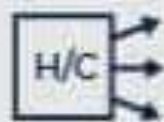


3. Ceiling Finishes

- Plaster Board
- Plaster Board cove moulding
- Steel Lathing
- Plastering
- Ceiling linings
- Ceiling tiles
- Spray plaster
- Ceiling papers
- Timber Cladding
- Expanded Polystyrene Tiles
- PVC Cladding
- Ceiling Painting



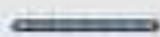
Heater/Cooler



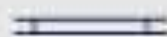
Heater/Cooler
Horizontal



Radiator



Convactor



Radiant Panel



Radiant Panel 2



Open Tank



Closed Tank



Water Surface



Pipe Coils



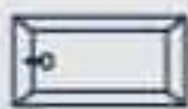
Sink



Basin



Toilet



Bath



Bath Side



Toilet (side)



Bath (side)



End View